

WHAT DO SEQUENTIAL GENERATIVE LATENTS ENCODE?

PROBING A DREAMERV3 WORLD MODEL AGAINST A STATIC VAE

Ricky (Xinrui) Bao
University of Chicago
rickybao@uchicago.edu

ABSTRACT

World models such as DreamerV3 learn sequential generative latents from streams of observations, while unsupervised representation learning is often studied on static data. We ask what this sequential latent encodes beyond a static, per-frame model. We freeze a DreamerV3 recurrent state-space model (RSSM) and a static frame-level VAE trained on the same proprioceptive Walker-Walk stream, then probe both with held-out ridge and MLP regressors across targets (current state, future state, gait phase), probe sites, and the static model’s temporal receptive field. In this near-fully observed proprioceptive setting, per-frame current state is decoded comparably well by both models; static VAE latents aggregated post-hoc over the past match and slightly exceed the RSSM at current-state decoding, while a few recent frames do not. These current-state results do not cover pixel or partially observed settings. For prediction, the RSSM prior reaches $R^2 \approx 0.50$ at a 20-step horizon, versus 0.16 for a posterior readout and 0.46 for learned dynamics in the VAE latent space; under the primary linear probe, the RSSM leads at every horizon, though the margin narrows. The recurrent posterior also makes gait phase linearly decodable. Finally, generative likelihood and probe R^2 decouple: the RSSM has lower reconstruction NLL than the VAE without a matching gain in linearly decodable state.

1 INTRODUCTION

A world model is a learned state representation that can be rolled forward to predict future states (Ha & Schmidhuber, 2018). DreamerV3 (Hafner et al., 2025) uses an RSSM (Hafner et al., 2019): observations enter a stochastic latent concatenated with a deterministic recurrent state, and the model reconstructs observations while predicting its next latent. Because world models are increasingly reused as pretrained representations for control, we ask what this sequential generative latent encodes beyond a static, per-frame encoder trained on the same observations.

We freeze a trained DreamerV3 RSSM and a static VAE with the same encoder and decoder architecture (trained from scratch, not shared weights), observation model, and training frames, so architecture and data are held fixed. We probe both with held-out regression (Alain & Bengio, 2016), varying target, probe site, and the static model’s temporal receptive field, and report held-out generative likelihood. In proprioceptive Walker-Walk, per-frame state content is shared, post-hoc aggregation closes the current-state gap, the RSSM prior retains higher future-state R^2 than a posterior readout and makes gait phase linearly decodable, and lower generative NLL does not imply a richer linear state code. Code and figures are released for reproducibility.¹

¹Code: <https://github.com/shalalababa/dreamerv3/tree/intrinsic-pretraining>; raw run outputs are omitted from the repository.

2 RELATED WORK

Probing representations. Linear and shallow probes measure how well a frozen representation supports a simple readout of a target attribute (Alain & Bengio, 2016). Because probe scores lower-bound decodable information, depend on the probe family, and are variance-weighted for vector targets, we fix the probe family, use ridge as primary and an MLP as a robustness check, and flag disagreements. Closest in spirit, ST-DIM introduced an Atari benchmark that evaluates unsupervised representations by recovery of ground-truth state variables (Anand et al., 2019); we use the same diagnostic logic for a narrower controlled RSSM-vs-VAE comparison.

World-model representations. World models learn latent states by predicting future observations (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; 2025); related objectives such as CPC also use future prediction as a representation signal (van den Oord et al., 2018). Reward-free world-model pretraining usually reports downstream return (Sekar et al., 2020; Liu & Abbeel, 2021; Laskin et al., 2021); we instead probe the frozen latent directly and compare it with a static VAE (Kingma & Welling, 2014; Higgins et al., 2017) sharing architecture and data.

3 BACKGROUND AND SETUP

RSSM world model. At each step, the RSSM keeps a deterministic state $h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$, a posterior $z_t \sim q_\phi(z_t | h_t, x_t)$ that sees the observation, and a prior $\hat{z}_t \sim p_\phi(\hat{z}_t | h_t)$ that does not. A decoder reconstructs $\hat{x}_t = d_\phi(h_t, z_t)$. Our z_t has 32 categorical variables with 4 classes each (128 dimensions), and h_t has 512 dimensions. Training uses reconstruction and a KL term aligning posterior and prior (Hafner et al., 2025).

Static VAE. The baseline removes recurrence and dynamics while holding encoder/decoder architecture, observation model, and training frames fixed. It maps a single frame to a diagonal-Gaussian latent and is trained on

$$\mathcal{L}_\beta(x) = \underbrace{\mathbb{E}_{q(z|x)}[-\log p(x | z)]}_{\text{reconstruction}} + \beta D_{\text{KL}}(q(z | x) \| \mathcal{N}(0, I)), \quad (1)$$

the negative ELBO for $\beta=1$ (Kingma & Welling, 2014; Higgins et al., 2017). Both models use the same unit-variance Gaussian observation model in symlog space, so held-out log-likelihoods are computed identically. The VAE latent has 128 dimensions to match the RSSM stochastic latent. A probe gap therefore reflects sequential structure and latent-family differences rather than the encoder, decoder, or data; remaining differences are revisited in the limitations.

Task, models, and data. We use DeepMind Control Walker-Walk (Tassa et al., 2018) with proprioceptive observations. We train three DreamerV3 seeds and one VAE per seed on that seed’s replay buffer; two seeds run to 500K environment steps and one to 1.1M. Probe figures show all seeds as points with the mean as bar or line. The 1.1M seed does not systematically dominate the probe scores, so we show it with the others; a possible training-length effect instead appears in generative likelihood, reported per seed (§5.5). Probe trajectories are 40 held-out episodes of 1000 steps from the frozen evaluation policy, with full MuJoCo physics state logged for targets.

4 PROBING METHODOLOGY

For a frozen representation z and a target y we fit a probe g and report the held-out coefficient of determination

$$R^2 = 1 - \frac{\sum (y - g(z))^2}{\sum (y - \bar{y}_{\text{train}})^2}, \quad (2)$$

against the train-set mean baseline; uninformative representations score ≈ 0 and held-out scores can be negative. For vector targets we report pooled, variance-weighted R^2 . We use the same ridge regressor and two-hidden-layer MLP at all sites, report ridge in the main text, and treat MLP as a robustness check; conclusions hold under both probes except where noted. Train and test splits are by episode. We vary:

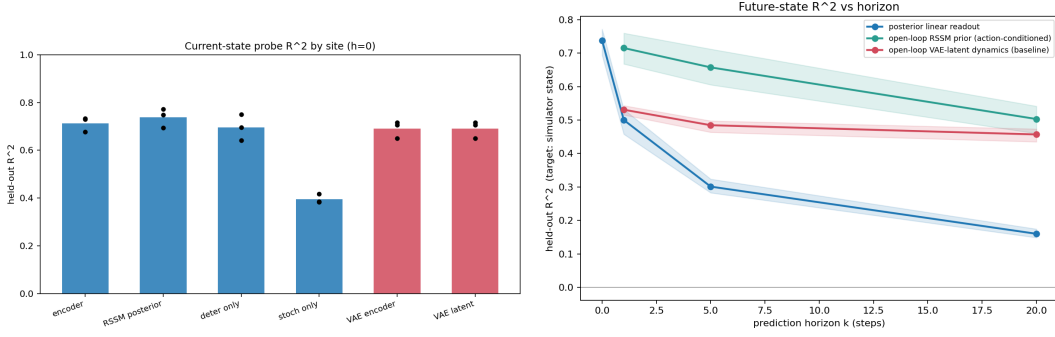


Figure 1: Held-out probe R^2 (points are seeds; bars/lines are means). **Left:** current state is decoded similarly by the encoder, RSSM posterior, RSSM deterministic state, and VAE; the stochastic latent alone is weaker. **Right:** the posterior readout decays quickly; the open-loop RSSM prior retains the most future-state R^2 under the primary linear probe, with VAE-latent dynamics a strong baseline.

- **Probe target:** current state; future state at $k \in \{1, 5, 20\}$; gait phase (encoded as $(\cos, \sin)$ of the analytic-signal phase of the antiphase hip oscillation); discounted return-to-go; and time-to-fall.
- **Probe site:** encoder output; RSSM posterior (h_t and $\mathbb{E}[z_t]$, also probed separately); RSSM prior, both one-step and open-loop k -step (roll the prior forward with the recorded actions and probe the imagined state); VAE encoder output; VAE latent mean; and the VAE latent rolled open-loop by a learned action-conditioned dynamics model (§5.2), the static analog of the RSSM prior.
- **Receptive field:** the VAE latent with a causal window. For $K \in \{1, 4, 16\}$ we concatenate recent latents; for $K=\text{all}$ we use the current latent plus the causal mean over the episode.

We additionally report held-out reconstruction and one-step predictive log-likelihood, computed identically for both models (§3).

5 RESULTS

5.1 PER-FRAME CURRENT STATE IS SHARED; WITHIN THE RSSM IT LIES MAINLY IN h_t

Figure 1 (left) shows that the encoder, RSSM posterior, RSSM deterministic state, VAE encoder, and VAE latent all decode current state at $R^2 \approx 0.69\text{--}0.74$. This content sits mainly in the RSSM deterministic state (≈ 0.70); the categorical stochastic latent alone is weaker (≈ 0.40 linear, ≈ 0.55 MLP). The deterministic state alone (0.70) is about level with the VAE latent (0.69); the posterior’s small advantage (0.74) comes from combining it with complementary information in the stochastic latent. At equal width, the continuous VAE latent is a stronger linear state code than the RSSM stochastic latent (0.69 versus 0.40). The MLP does not favor the RSSM either, consistent with per-frame content being shared.

5.2 PREDICTIVE CONTENT: OPEN-LOOP PRIOR VERSUS POSTERIOR READOUT

A linear readout of future state from the posterior decays sharply ($R^2 = 0.74, 0.50, 0.30, 0.16$ at $k = 0, 1, 5, 20$; Figure 1, right). Rolling the prior forward with recorded actions retains higher R^2 (0.72, 0.66, 0.50 at $k = 1, 5, 20$); at 20 steps this is about three times the posterior-readout R^2 (0.50 versus 0.16), and the MLP shows the same prior-vs-posterior pattern (0.43 versus about zero). To test whether this is only the benefit of having dynamics, we fit a post-hoc VAE-latent forward model $g(z_t, a_t) \approx z_{t+1} - z_t$ on probe train episodes and roll it open-loop. This baseline is asymmetric by design: the RSSM transition was co-trained with its latent, whereas VAE dynamics are fit after representation learning on frozen latents. Even so, the baseline recovers much more future state than a static latent readout (0.53, 0.49, 0.46 versus 0.32, 0.19, 0.11). Under the primary linear probe, the RSSM prior still leads within every seed (0.72, 0.66, 0.50 versus 0.53, 0.49, 0.46): the unrounded

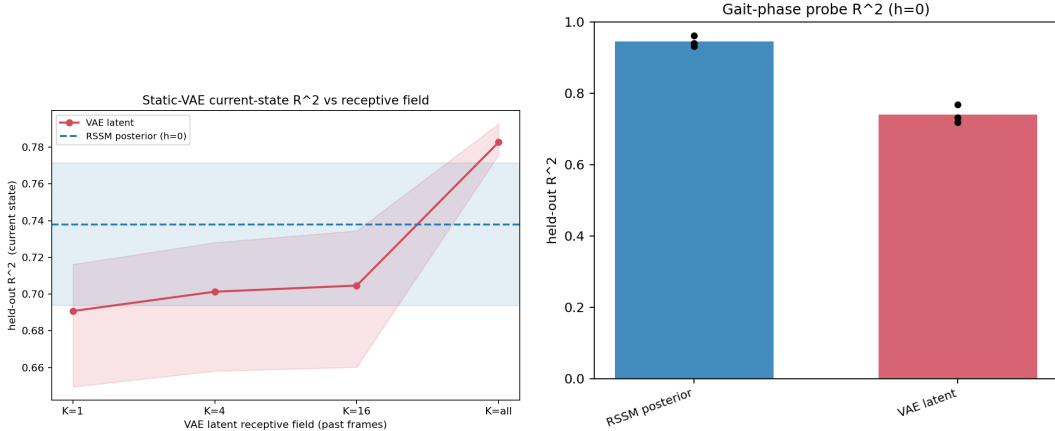


Figure 2: **Left:** VAE current-state R^2 rises with temporal receptive field; only whole-episode aggregation overtakes the RSSM posterior reference. **Right:** gait phase has higher linear-probe R^2 under the recurrent posterior than under the single-frame VAE latent.

mean margin shrinks from $+0.18$ at $k = 1$ (seed range $+0.15$ to $+0.22$) to $+0.05$ at $k = 20$ (range $+0.02$ to $+0.07$). Under the MLP, the RSSM and VAE-dynamics models are separated at $k = 1$ and $k = 5$ but essentially tied at $k = 20$, so the head-to-head advantage is strongest in the linear-probe setting. Both action-conditioned models approach similar long-horizon accuracy as open-loop error compounds (Appendix A, Figure 3); the RSSM advantage is sharpest at short horizons, where its co-trained transition is most accurate.

5.3 POST-HOC TEMPORAL AGGREGATION OF STATIC FEATURES CLOSES THE CURRENT-STATE GAP

Figure 2 (left) shows VAE current-state R^2 rising from 0.69 (one frame) through 0.70 and 0.71 (the most recent 4 and 16 frames) to 0.78 with the causal whole-episode mean, overtaking the RSSM posterior (0.74). Concatenating a handful of recent frames is not enough; only whole-episode aggregation closes and reverses the gap. Thus the current-state content the RSSM exposes is recoverable from static features by a hand-specified temporal summary, not learned recurrence.

5.4 GAIT PHASE: THE RSSM MAKES A CYCLIC QUANTITY LINEARLY DECODABLE

Figure 2 (right) shows the largest linear RSSM–VAE gap: gait phase is decoded at $R^2 \approx 0.95$ from the posterior versus ≈ 0.74 from the VAE latent ($+0.21$; seed ranges 0.93–0.96 versus 0.72–0.77). Phase is position within a cycle, which a single frame cannot disambiguate from a linear readout. A nonlinear MLP recovers much of this from the static latent (0.89 versus 0.95), so the RSSM mainly makes cyclic information more linearly accessible. Return-to-go is near zero for both models, and time-to-fall is unreliable due to censoring (Appendix C).

5.5 GENERATIVE QUALITY AND PROBE QUALITY DECOUPLE

At matched training (Table 1), the RSSM reconstructs the current frame with lower NLL than the VAE (0.62–0.67 versus 1.77–1.95 for the two 500K seeds, and lower per-frame NLL on 90–97% of frames), yet its current-state probe R^2 is only marginally higher. Seed 0, trained longer, has still lower NLL (0.36) but probe R^2 within seed noise. Reconstruction likelihood and linear-probe decodability therefore agree only partially.

6 DISCUSSION

In this near-fully observed proprioceptive Walker setting, the sequential latent does not encode more of the current observable state than a static VAE with a hand-specified causal summary of past

Table 1: Held-out NLL (lower is better), mean over the held-out probe set. The RSSM reconstructs better than the VAE, especially for the longer-trained seed 0, without a corresponding current-state probe gain.

Held-out NLL	seed 0 (1.1M)	seed 1 (500K)	seed 2 (500K)
RSSM posterior reconstruction	0.36	0.67	0.62
RSSM prior one-step prediction	0.80	1.78	1.47
Static VAE reconstruction	1.58	1.95	1.77

latents. Its advantages are predictive and coordinate-shaping: the action-conditioned prior forecasts better than a posterior readout under both probe families, and under the primary linear probe it also beats the VAE-dynamics baseline in all seeds (0.50 versus 0.46 at 20 steps), although the MLP comparison ties at that horizon. The RSSM also makes gait phase linearly decodable, while a nonlinear probe recovers most of it from the static latent. Separately, lower reconstruction NLL does not yield a matching gain in linear state decodability, so likelihood and probe R^2 should not be treated as interchangeable measures of representation quality. Practically, when only current-state decoding is needed, a static VAE with simple causal pooling is enough; sequential pretraining is most useful for multi-step prediction and for making cyclic structure linearly accessible.

Limitations. The comparison still differs in latent family, prior, objective, and probed dimensionality (640 versus 128); matched-width probes control only part of this. The VAE-dynamics baseline is a post-hoc model trained on frozen VAE latents from probe episodes, whereas the RSSM transition was co-trained with its representation; stronger VAE dynamics could narrow the long-horizon gap. We report pooled, variance-weighted R^2 , leaving per-component probes to future work. Finally, we study one domain, one model size, and proprioception, where a single frame nearly determines current state. Pixel and other partially observed regimes remain untested, and only one seed ran to 1.1M steps, so the training-length effect on NLL is suggestive.

ACKNOWLEDGMENTS

Built on the open-source DreamerV3 implementation (Hafner et al., 2025). Compute was provided by the UChicago Research Computing Center.

REFERENCES

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.
- Ankesh Anand, Evan Racah, Sherjil Ozair, Yoshua Bengio, Marc-Alexandre Côté, and R Devon Hjelm. Unsupervised state representation learning in Atari. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems (NeurIPS)*. 2018.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning (ICML)*, 2019.
- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025.
- Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed, and Alexander Lerchner. beta-VAE: Learning basic visual concepts with a

constrained variational framework. In *International Conference on Learning Representations (ICLR)*, 2017.

Diederik P Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations (ICLR)*, 2014.

Michael Laskin, Denis Yarats, Hao Liu, Kimin Lee, Albert Zhan, Kevin Lu, Catherine Cang, Lerrel Pinto, and Pieter Abbeel. URLB: Unsupervised reinforcement learning benchmark. In *NeurIPS Datasets and Benchmarks Track*, 2021.

Hao Liu and Pieter Abbeel. Behavior from the void: Unsupervised active pre-training. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Ramanan Sekar, Oleh Rybkin, Kostas Daniilidis, Pieter Abbeel, Danijar Hafner, and Deepak Pathak. Planning to explore via self-supervised world models. In *International Conference on Machine Learning (ICML)*, 2020.

Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind control suite. *arXiv preprint arXiv:1801.00690*, 2018.

A ron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

A OPEN-LOOP PREDICTION

Figure 3 shows a representative open-loop rollout (seed 0): a 25-step posterior context followed by 75 steps of prior imagination using the recorded actions, decoded back to observation units. The posterior tracks the truth closely; the prior then carries the gait oscillations forward for many steps before gradually drifting, and the pooled prediction RMSE grows smoothly with horizon.

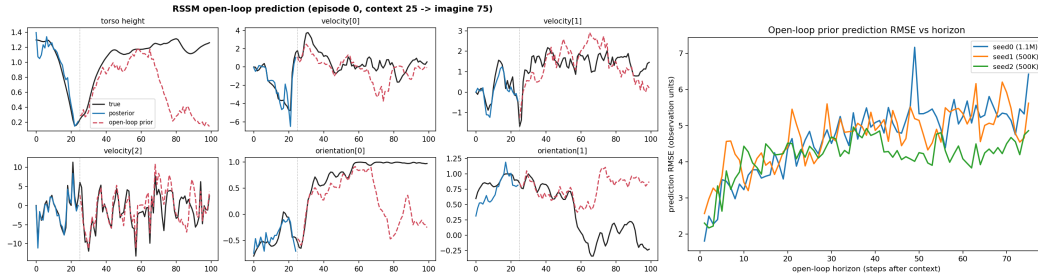


Figure 3: Open-loop prediction. Left: true (black), posterior reconstruction (solid), open-loop prior prediction (dashed) per observation variable. Right: pooled open-loop prediction RMSE versus horizon, per seed.

B GENERATIVE QUALITY AND RECONSTRUCTION ADVANTAGE

Figure 4 is the visual companion to Table 1: held-out NLL by quantity and seed (left), and the fraction of frames on which the RSSM reconstructs better than the VAE (right).

C NULL AND UNRELIABLE TARGETS

Two derived targets are reported for completeness but excluded from the main findings. Return-to-go (discounted future reward) has near-zero R^2 for both models (≈ -0.01 posterior, ≈ 0 VAE) on Walker-Walk, where per-step reward is nearly constant and not linearly decodable. Time-to-fall is highly seed-dependent (posterior R^2 of 0.06, 0.50, and 0.98 across seeds) because a competent walker rarely falls, so the target is dominated by rare, heavily right-censored events; we therefore do not draw conclusions from it.

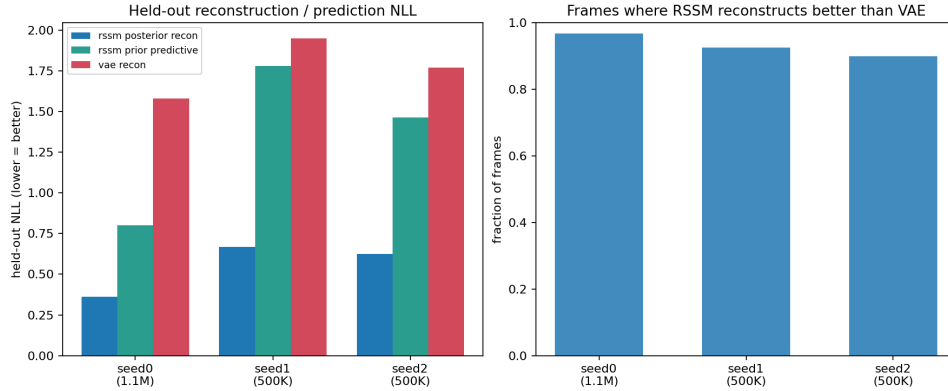


Figure 4: Held-out reconstruction and prediction NLL (left) and the fraction of frames where the RSSM reconstructs better than the VAE (right). Seed 0 (1.1M steps) has much lower NLL than the 500K seeds, while its probe R^2 does not correspondingly increase.

D REPRODUCIBILITY DETAILS

DreamerV3 uses the `dmc_proprio/size1m` configuration ($h_t \in \mathbb{R}^{512}$, z_t a set of 32×4 categorical, encoder and decoder MLPs of width 64, depth 3, symlog preprocessing). The VAE uses the same encoder and decoder architecture, with a 128-dimensional Gaussian latent, $\beta=1$, trained from scratch for 60K steps (Adam, lr 3×10^{-4}) on each seed’s replay buffer. Probes use ridge (regularization selected on a validation split) and a two-hidden-layer, width-256 ReLU MLP. The VAE-latent dynamics baseline is a two-hidden-layer width-256 residual MLP trained with a curriculum multi-step rollout loss (the rollout length ramped $1 \rightarrow 20$) for 6000 steps (Adam, lr 10^{-3} , gradient clip 1.0) on the probe train episodes. The probe set is 40 held-out episodes of 1000 steps from the frozen evaluation policy with logged physics state. All figures in `final_report/iclr2026/figures/` were built from the per-seed outputs.